BIOCHEMISTRY CUME

March 14, 2014

Environmental Aspects of Protein Structure and Function

1. What is the net charge on the following peptide at pH 4; pH 7, and pH 9? Please show sufficient work to justify your answers. (30 pts.)

HACYRRKKGLFHHHR

2. At pH of 4, is the peptide above likely to be structured as a random coil or helical? Briefly explain your rationale. (20 pts.)

3. Briefly describe what an intrinsically disordered protein is. (25 pts.)

4. It has been shown that "scrambled" ribonuclease spontaneously converts to the fully active, native ribonuclease in the presence of trace amounts of β-mercaptoethanol in the protein solution. Explain why this result occurs. Based on your answer, deduce and explain the meaning of "scrambled," in the context of this problem. (25 pts.)